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UKUDLA Data Science Certificate

Version Control and Collaboration using Git & GitHub

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Why version a data science project?

```
analysis.R  
analysis-new.R  
analysis-final.R  
analysis-final-revised.R  
analysis-final-revised-2.R
```

Version control provides

- one authoritative project;
- a traceable sequence of changes;
- meaningful recovery points;
- parallel and reviewable work; and
- a link between decisions and results.

Saving a file is not the same as recording a version. Version control turns a folder of files into a project with a documented history.

Why this matters for food-systems data science

Food-systems analyses combine many changing elements:

- data-acquisition and cleaning decisions;
- definitions, units, study units, and time periods;
- statistical code and model specifications;
- figures, reports, and interpretation; and
- software-environment descriptions.

A small change in a filtering rule or unit conversion can change the scientific conclusion.
Git helps connect that conclusion to a specific, inspectable state of the project.

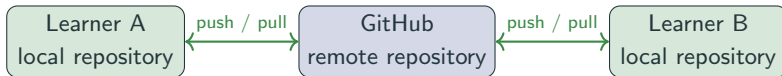
Git and GitHub: local history and shared collaboration

Git

- Distributed version control system
- Runs on the local computer
- Records project snapshots
- Most operations work offline

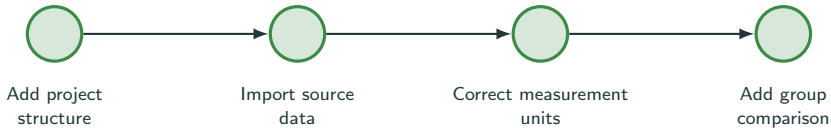
GitHub

- Online platform hosting Git repositories
- Provides a shared remote location
- Adds review and access control
- Supports issues, pull requests, and automation



A commit is a meaningful project snapshot

A commit records selected changes with an author, timestamp, message, and connection to the preceding project history.



- A commit should contain one coherent change.
- The diff records *what* changed.
- The message should help explain *why*.

From an edited file to shared history



Workflow

- Editing changes the working directory.
- Staging selects what belongs to the next commit.
- Committing records a local snapshot.
- Pushing shares local commits with the remote repository.

Not everything should be tracked!

- secrets and credentials;
- confidential data;
- large downloaded data;
- generated outputs;
- package libraries and caches; and
- editor-specific state.

Inspect before you record

```
git status
git diff
git add path/to/file
git diff --staged
git commit -m "Correct measurement unit conversion"
git push
```

Questions before committing

- Do I understand each change?
- Do these changes belong together?
- Did I include generated or private files?

A useful commit

- is small and coherent;
- leaves the project usable; and
- explains why the change matters.

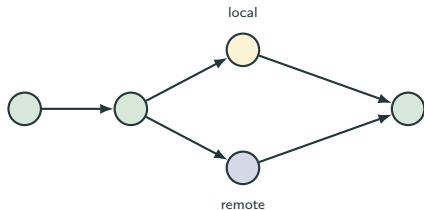
A collaborative Git cycle

`inspect → pull → edit → commit → push`

- Pull remote changes before beginning a new task.
- Make small, focused changes.
- Communicate which files or tasks people are working on.
- Push completed commits so collaborators can build on them.
- Integrate frequently; long-lived divergence increases effort.

Collaboration means exchanging and integrating commits—not emailing copies of “final” files.

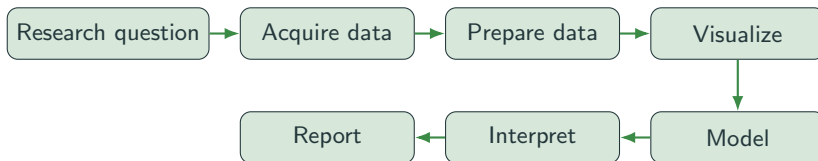
Divergence, merging and scientific judgment



- Diverging histories are normal.
- Git often combines different changes automatically.
- A conflict asks a human to choose or integrate.
- The integrated analysis must still run and retain its intended meaning.

Git identifies overlapping text. It cannot decide which research assumption, data transformation, or interpretation is scientifically correct.

Version control supports the complete data science workflow



Git and GitHub

- mark meaningful stages;
- connect code changes to results; and
- support sharing and review.

Track deliberately

- Track code, documentation, and lockfiles.
- Exclude secrets, sensitive data, caches, and unsuitable large files.

Key message: version meaningful project states

1. Git records meaningful states of a project.
2. GitHub provides a shared location and collaboration tools.
3. Inspecting changes is part of responsible data science.
4. Small commits and frequent synchronization reduce problems.
5. Version control supports reproducibility, but does not replace data, environment, and workflow documentation.

Detailed setup, repository, workflow, and troubleshooting instructions are provided in the learning-module guides.

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